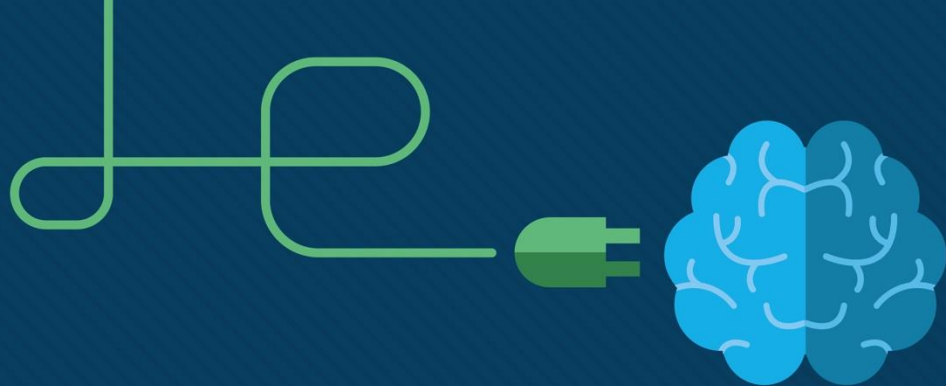




Module 2: Switching Concepts

Switching, Routing, and
Wireless Essentials v7.0
(SRWE)



Module Objectives

Module Title: Switching Concepts

Module Objective: Explain how Layer 2 switches forward data.

Topic Title	Topic Objective
Frame Forwarding	Explain how frames are forwarded in a switched network.
Switching Domains	Compare a collision domain to a broadcast domain.

2.1 Frame Forwarding

Switching in Networking

Two terms are associated with frames entering or leaving an interface:

- **Ingress** – entering the interface
- **Egress** – exiting the interface

A switch forwards based on the ingress interface and the destination MAC address.

A switch uses its MAC address table to make forwarding decisions.

Note: A switch will never allow traffic to be forwarded out the interface it received the traffic.



Port Table

Destination Addresses	Port
EE	1
AA	2
BA	3
EA	4
AC	5
AB	6

The Switch Learn and Forward Method

The switch uses a two step process:

Step 1. Learn – Examines Source Address

- Adds the source MAC if not in table
- Resets the time out setting back to 5 minutes if source is in the table

Step 2. Forward – Examines Destination Address

- If the destination MAC is in the MAC address table it is forwarded out the specified port.
- If a destination MAC is not in the table, it is flooded out all interfaces except the one it was received.

Switch Forwarding Methods

Switches use software on application-specific-integrated circuits (ASICs) to make very quick decisions.

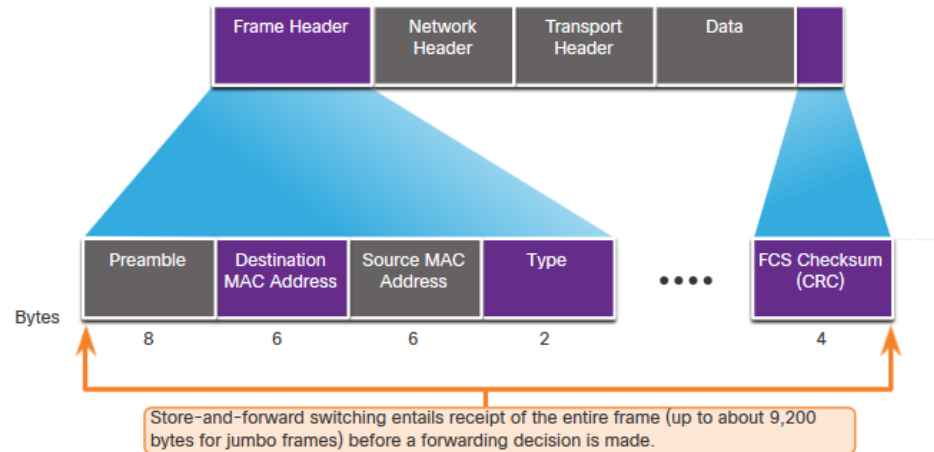
A switch will use one of two methods to make forwarding decisions after it receives a frame:

- **Store-and-forward switching** - Receives the entire frame and ensures the frame is valid. Store-and-forward switching is Cisco's preferred switching method.
- **Cut-through switching** – Forwards the frame immediately after determining the destination MAC address of an incoming frame and the egress port.

Store-and-Forward Switching

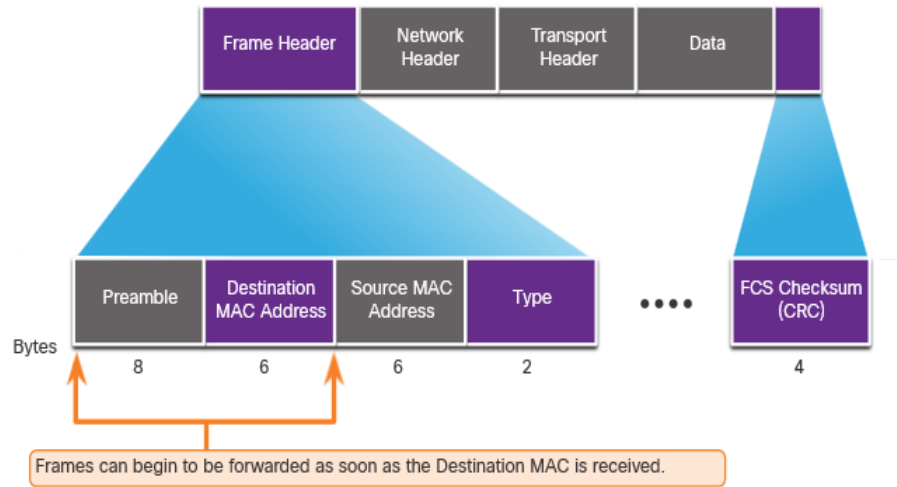
Store-and-forward has two primary characteristics:

- Error Checking – The switch will check the Frame Check Sequence (FCS) for CRC errors. Bad frames will be discarded.
- Buffering – The ingress interface will buffer the frame while it checks the FCS. This also allows the switch to adjust to a potential difference in speeds between the ingress and egress ports.



Frame Forwarding

Cut-Through Switching



- Cut-through forwards the frame immediately after determining the destination MAC.
- Fragment (Frag) Free method will check the destination and ensure that the frame is at least 64 Bytes. This will eliminate runs.

Concepts of Cut-Through switching:

- Is appropriate for switches needing latency to be under 10 microseconds
- Does not check the FCS, so it can propagate errors
- May lead to bandwidth issues if the switch propagates too many errors
- Cannot support ports with differing speeds going from ingress to egress